AI Stock Predictor

# User Guide & Technical Overview

Rank the S&P 500, forecast risk, and see how well the models really do.

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## 1 Introduction

Welcome to your personal stock research dashboard. It scores every company in the S&P 500, forecasts how large each stock's moves are likely to be, and then **grades its own predictions** against simple baselines on data the models never saw.

Everything runs on your own computer. Data is downloaded once from free public sources and kept on disk, so after the first update the app works offline, with no accounts or API keys.

| 499 | 10 yrs | 4 | $0 |
|---|---|---|---|
| stocks ranked | of daily prices | macro series | paid services |

**An honest tool.** Forecasting the direction of daily stock returns is extremely hard, and the Model Report page says so plainly when the model has no edge. What does work is forecasting *risk*: how big a stock's moves will be. That is the part to rely on.

## Key features

### Markets watchlist

499 stocks sorted by market cap, with company logos, 40-day trend lines, search, sector filter and sortable columns.

### Risk Forecast

Every stock is rated Calm, Normal or Wild, with the typical 5-day swing, forecast by a volatility model.

### Stock pages

Candlestick charts from 1 month to 5 years, with the predicted price, a typical-swing band and a 52-week range.

### Macro page

10-year yield, yield curve, credit spread and the Fed funds rate from FRED, with a plain-English note on each.

### Model Report

The scorecard: accuracy against "always guess up", week-by-week ranking skill, and the risk forecast's score.

### Private and accessible

Local only, keyboard friendly, screen-reader labels, and an Animations switch that follows your system setting.

# 2 How to use the application

Using the app is a short process: install, start, update once, then explore.

### 1 Install and start

In the project folder run these two commands. The app opens in your browser at `http://127.0.0.1:8000`.

```
$ pip install -r requirements.txt
$ python server.py
```

### 2 Enter the app

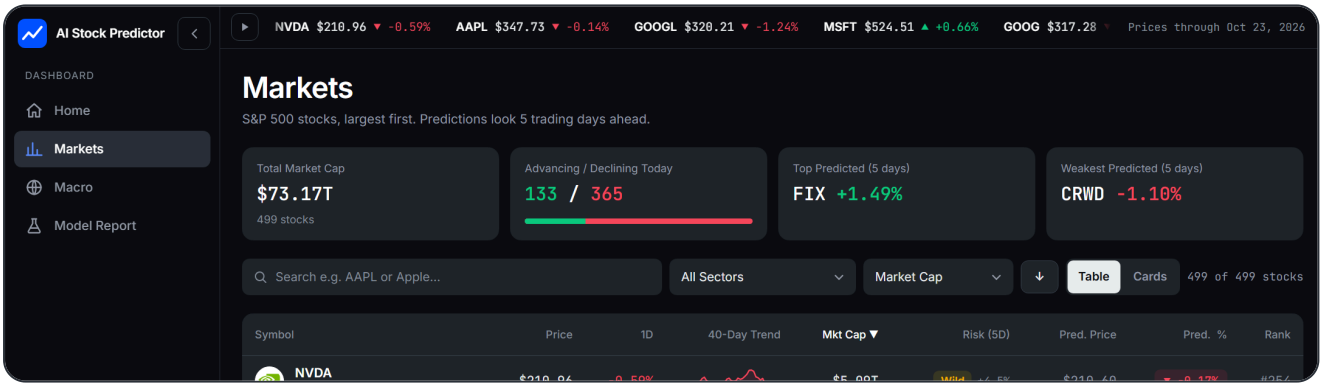
A loading bar tracks the real start-up steps. At 100% select **Enter App**, then **Open Dashboard** on the landing page.

### 3 Update once

In the sidebar select **Update Data & Retrain**. This is the only step that needs the internet. It downloads about 10 years of prices for ~500 stocks, share counts, logos and the FRED macro series, then trains both models. The first run takes a couple of minutes.

### 4 Explore the Markets page

Click a column header to sort (default: market cap, largest first), search, filter by sector, or switch between **Table** and **Cards**. The **Risk (5D)** column shows the rating and typical swing. Select a stock to open its page.




The Markets page: ticker tape, summary tiles, filters and the market-cap watchlist with risk tags.

## 2.1 Reading a stock page

- **Chart.** Candlesticks, volume and 20- and 50-day averages. Hover, or focus the chart and use the arrow keys, to read prices.
- **Predicted price and swing band.** The dashed blue line is the prediction; the shaded band is the typical 5-day swing. The prediction sits well inside it, which is the honest picture of the uncertainty.
- **Side panels.** Expected change, rank, risk rating, market cap and the 52-week range.

## 2.2 Handy controls

- **Sidebar.** Collapse it with the arrow button. **Animations** turns motion on or off (it starts from your system setting).
- **Ticker.** The top bar scrolls the largest companies and macro readings. Pause/Play stops it.

# 3 Technical overview

The app is a small Python pipeline behind a local web server. Every layer uses free, open-source tools.

## 3.1 The brain: two simple models

Both models are **Ridge regressions** from scikit-learn, trained on every stock together (about 940,000 stock-days). Simple models were chosen on purpose: they overfit less, and in testing a more complex model did not do better.

- **Return model.** Estimates each stock's return over the next 5 trading days from 20 inputs: returns over 1 to 60 days, moving-average gaps, RSI, volume change, volatility, strength versus the market and sector peers, rank among all stocks, and four FRED macro readings.
- **Risk model.** Forecasts next-week volatility from recent volatility over 5, 20 and 60 days, today's absolute move, market-wide volatility and the stock's volatility relative to the market.

## 3.2 The data engine

**yfinance** supplies adjusted daily prices and share counts (market cap is computed locally as shares × price). **FRED** supplies rates and credit series through its public CSV downloads, and **Wikipedia** supplies the S&P 500 list and sectors. Downloads are batched with retries, corrupt rows are dropped, stale tickers are excluded, and a data-quality report flags gaps and extreme one-day moves.

## 3.3 The server

**FastAPI** running on **Uvicorn** exposes the data as JSON on `127.0.0.1` only. Updates run as a background job, responses are gzip-compressed, and finished predictions are saved to disk so a restart takes seconds.

## 3.4 The interface

Plain HTML, CSS and JavaScript (no framework). Charts are drawn on a **canvas**, trend lines are **SVG**, numbers and dates use the browser's **Intl** formatting, and the type is **Inter** and **JetBrains Mono** served from local files.

| Layer | Technology | Used for |
|---|---|---|
| **Data** | yfinance, FRED, Wikipedia, requests, lxml | Prices, share counts, macro series, index list |
| **Python** | Python 3.11, pandas, NumPy, joblib | Cleaning, features, saving trained models |
| **Machine learning** | scikit-learn (Ridge) | Return model and risk model |
| **Server** | FastAPI NEW Uvicorn NEW | JSON API, background updates, static files |
| **Interface** | HTML, CSS, JavaScript NEW | Watchlist, pages, ticker, preloader |
| **Graphics** | Canvas, SVG, Inter, JetBrains Mono NEW | Charts, sparklines, typography |

**Retired since version 1:** Random Forest per stock, CustomTkinter, mplfinance, Matplotlib and Pillow.

# 4 Results, reported honestly

Both models are scored on the last two years of data they never trained on, in a time-ordered test with no peeking at the future.

RETURN FORECAST

## No reliable signal

| | |
|---|---|
| Direction right | 52.4% |
| Always guessing up | 52.3% |
| Best vs worst fifth, t-value | 0.01 |
| Top fifth beats the median | 50.6% |
| Weeks tested | 101 |

RISK FORECAST

## Real, measurable skill

| | |
|---|---|
| Variation explained | 23.2% |
| “Same as lately” baseline | 18.5% |
| Correlation | 0.482 |
| Baseline correlation | 0.447 |




Average daily move afterwards, by rating

### How to read this

- **Returns.** The model’s direction calls are right about as often as always guessing up, and its ranking of stocks is no better than chance (a t-value near 0; clear skill needs above 2). Treat the rank as a hint, never a signal.
- **Risk.** Stocks rated Wild really did move about twice as much as those rated Calm. Most of that is volatility persisting; the model adds a measurable gain over the simple baseline.
- **Size, not direction.** A Wild rating means big moves either way. It says nothing about whether a stock will go up or down.

## 5 Disclaimer: for educational purposes only

This application is an educational tool that demonstrates machine learning on financial data. **It is not financial advice**, and it does not place trades. The forecasts are based only on historical prices and public macro data, and they do not account for:

- Breaking news, world events or market sentiment, including unexpected “black swan” events.
- Company fundamentals such as earnings, debt and guidance.
- Transaction costs, taxes, liquidity and slippage.
- Delayed, revised or incorrect data from the free public sources it relies on.

All investing carries risk, including the loss of everything you invest. Always do your own research and consult a qualified financial adviser before making any investment decision.